Assignment 3 Time Series Weather Forecasting

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Objective

The objective was to improve a 24-hour temperature forecast for the Jena weather data by comparing recurrent neural network designs. I used validation mean absolute error (MAE) to select the final model because this is a regression problem and a lower MAE means the forecast is closer to the observed temperature.

Main Result

The best model was a 32-unit LSTM with 0.25 recurrent dropout and 0.50 output dropout. It reached a validation MAE of 2.31 degrees C and a test MAE of 2.49 degrees C. The test error was 5.1% lower than the naive forecast.

Experiment Design

Item	Choice
Dataset	Jena climate observations from 2009 through 2016 with 14 weather measures
Input window	Five days sampled hourly, giving 120 time steps
Prediction	Temperature 24 hours after the end of the input window
Data split	First 50% training, next 25% validation, final 25% test
Preprocessing	Standardized with the training-set mean and standard deviation
Selection rule	Lowest validation MAE; test set used only after choosing the winner

Models Tested

The six neural networks tested the specific changes suggested in the assignment: GRU unit count, GRU versus LSTM, recurrent dropout, stacked recurrent layers, and a Conv1D layer followed by a GRU. All models used RMSprop, mean squared error for the training loss, and early stopping based on validation MAE.

Validation Results

Table 1 summarizes the best validation epoch for each design. Only the selected model was evaluated on the test set, so the other test entries are left blank.

Model	Design	Best Epoch	Validation MAE	Test MAE
Naive baseline	Current temperature	-	2.44	2.62
GRU 16	Single GRU	2	2.39	-
GRU 32	Single GRU	2	2.42	-
LSTM 32	Single LSTM	1	2.46	-
LSTM 32 with dropout	LSTM, recurrent and output dropout	5	2.31	2.49
Stacked GRU 32 and 16	Two GRU layers with dropout	4	2.35	-
Conv1D plus GRU 32	Conv1D, pooling, then GRU	1	2.42	-

Table 1. Model comparison in degrees Celsius. Lower MAE is better.

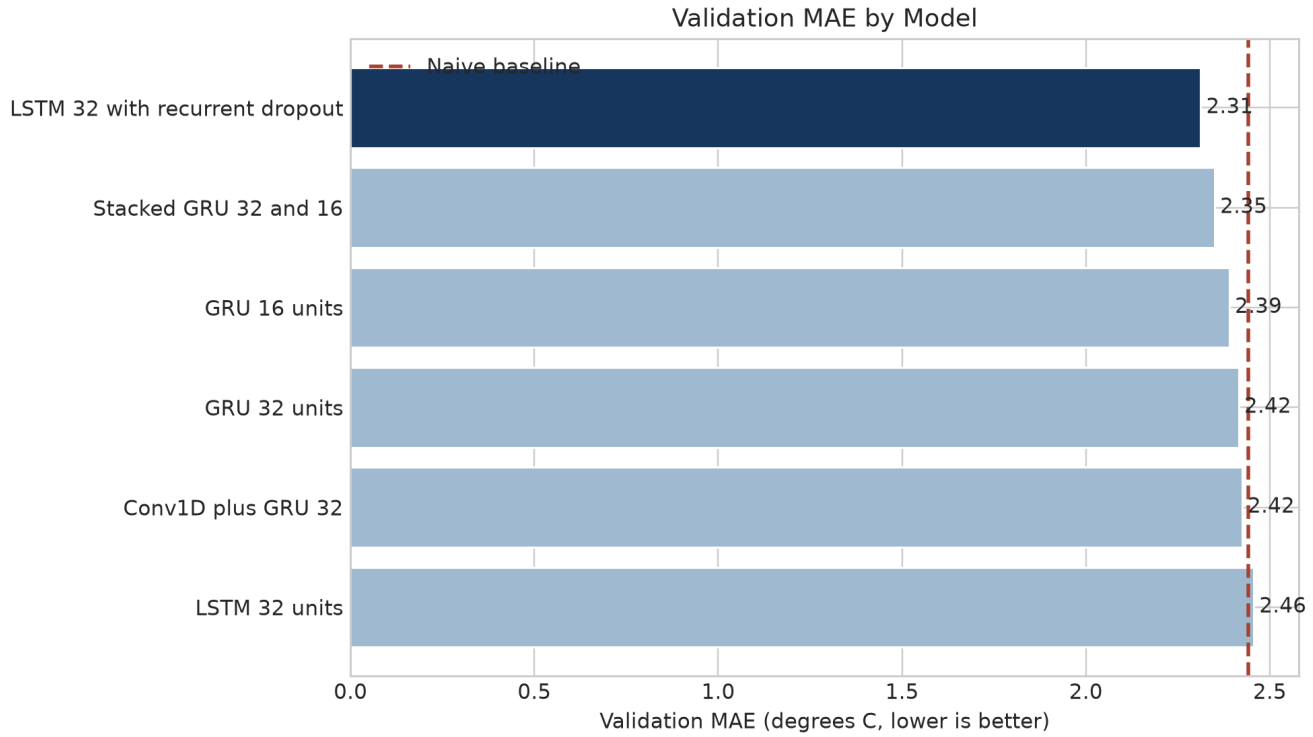


Figure 1. Validation MAE for the six neural networks and the naive baseline.

What Changed

The 16-unit GRU beat the naive validation MAE, but increasing it to 32 units made the result slightly worse. A plain 32-unit LSTM was also worse than the smaller GRU. Adding recurrent and output dropout to the LSTM reduced validation MAE to 2.31 degrees C, a 5.3% improvement over the naive validation forecast. The stacked GRU ranked second, but it required more training time and did not beat the regularized LSTM.

Final Model

The regularized LSTM reached its lowest validation MAE at epoch 5. Training MAE was still higher than validation MAE because dropout is active during training but not during validation. The validation curve leveled off after epoch 5, so additional training did not provide a meaningful improvement.

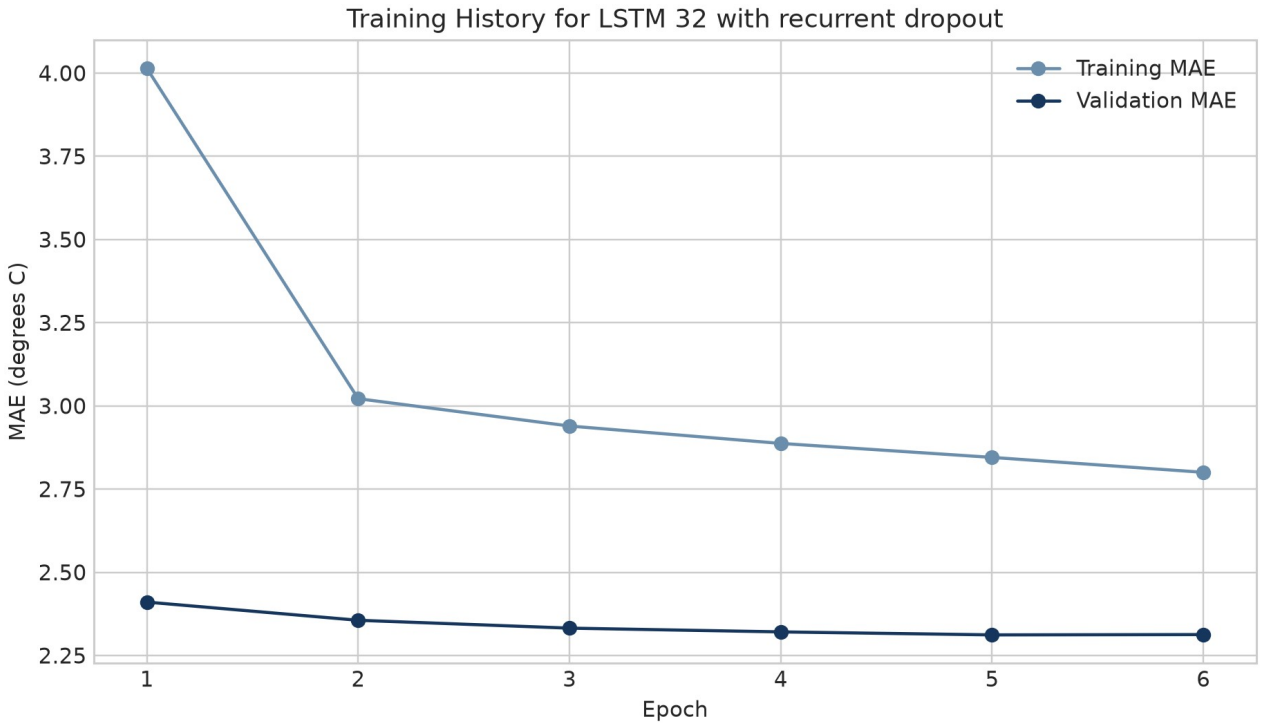


Figure 2. Training and validation MAE for the selected LSTM model.

Test Evaluation

After selecting the model from validation performance, I evaluated it once on the test set. The LSTM test MAE was 2.49 degrees C. The naive test MAE was 2.62 degrees C, so the model reduced the average absolute error by 0.13 degrees C, or 5.1%.

Conclusion

The experiment showed that more capacity did not automatically improve the forecast. The larger GRU and plain LSTM both underperformed the 16-unit GRU. The strongest improvement came from regularizing the LSTM with recurrent dropout and output dropout. The Conv1D plus GRU design also failed to improve the result, which is consistent with the importance of preserving sequence order for this forecasting problem. Based on the validation and test results, the regularized LSTM is the best final model from the configurations tested.